

Assignment 3: Data Exploration

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on Data Exploration.

Directions

1. Rename this file `<FirstLast>_A03_DataExploration.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Assign a useful **name to each code chunk** and include ample **comments** with your code.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.
7. After Knitting, submit the completed exercise (PDF file) to the dropbox in Canvas.

TIP: If your code extends past the page when knit, tidy your code by manually inserting line breaks.

TIP: If your code fails to knit, check that no `install.packages()` or `View()` commands exist in your code.

Set up your R session

1. Load necessary packages (tidyverse, lubridate, here), check your current working directory and upload two datasets: the ECOTOX neonicotinoid dataset (`ECOTOX_Neonicotinoids_Insects_raw.csv`) and the Niwot Ridge NEON dataset for litter and woody debris (`NEON_NIWO_Litter_massdata_2018-08_raw.csv`). Name these datasets “Neonics” and “Litter”, respectively. Be sure to include the sub-command to read strings in as factors.

```
#import libraries
library(tidyverse); library(lubridate); library(here)

#check workspace
here()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```

#upload ECOTOX
Neonics <- read.csv(
  file = here('Data', 'Raw', 'ECOTOX_Neonicotinoids_Insects_raw.csv'),
  stringsAsFactors = T
)

#upload NEON
Litter <- read.csv(
  file = here('Data', 'Raw', 'NEON_NIWO_Litter_massdata_2018-08_raw.csv'),
  stringsAsFactors = T
)

```

Learn about your system

2. The neonicotinoid dataset was collected from the Environmental Protection Agency's ECOTOX Knowledgebase, a database for ecotoxicology research. Neonicotinoids are a class of insecticides used widely in agriculture. The dataset that has been pulled includes all studies published on insects. Why might we be interested in the ecotoxicology of neonicotinoids on insects? Feel free to do a brief internet search if you feel you need more background information.

Answer: We may want to know what insects the neonicotinoids effect. The farmer might know that it kills/deters some specific kind of insect that harms their crop, but how does the insecticides affect other insects that are not destructive to the crop? These other insects may be necessary to the function to the ecosystem. We would want to understand these affects of neonicotinoids in case it's use needs to be limited given its harmful affect on a vital insect.

3. The Niwot Ridge litter and woody debris dataset was collected from the National Ecological Observatory Network, which collectively includes 81 aquatic and terrestrial sites across 20 ecoclimatic domains. 32 of these sites sample forest litter and woody debris, and we will focus on the Niwot Ridge long-term ecological research (LTER) station in Colorado. Why might we be interested in studying litter and woody debris that falls to the ground in forests? Feel free to do a brief internet search if you feel you need more background information.

Answer: Forest litter and woody debris are needed for nutrient cycling, which supplies the soil with nitrogen and carbon. These nutrients help maintain forest health. Additionally, litter and debris can impact erosion, water paths, and forest fire patterns.

4. How is litter and woody debris sampled as part of the NEON network? Read the NEON_Litterfall_UserGuide.pdf document to learn more. List three pieces of salient information about the sampling methods here:

Answer: 1. Litter and wood debris size are both defined as butt end diameter <2cm and a length <50 cm but they are collected using different contraptions. 2. Samples were taken from sites that have woody vegetation of at least 2m tall. 3. Ground traps were sampled once per year while elevated traps were sampled more frequently, depending on the type of forest (deciduous or evergreen forests)

Obtain basic summaries of your data (Neonics)

5. What are the dimensions of the dataset?

```
dim(Neonics)
```

```
## [1] 4623 30
```

6. Using the `summary` function on the “Effect” column, determine the most common effects that are studied. Why might these effects specifically be of interest? [Tip: The `sort()` command is useful for listing the values in order of magnitude...]

```
sort(summary(Neonics$Effect)) #summary returns all values and their counts.
```

```
##      Hormone(s)      Histology      Physiology      Cell(s)
##           1           5           7           9
## Biochemistry Accumulation Intoxication Immunological
##          11          12          12          16
## Morphology      Growth      Enzyme(s)      Genetics
##          22          38          62          82
## Avoidance      Development Reproduction Feeding behavior
##          102          136          197          255
## Behavior      Mortality      Population
##          360          1493          1803
```

```
#Sort sorts these values from lowest to highest.
```

Answer: Population and mortality are the most common effects that are studied. This shows that the insecticide is having an effect on the insects presence in an area.

7. Using the `summary` function, determine the six most commonly studied species in the dataset (common name). What do these species have in common, and why might they be of interest over other insects? Feel free to do a brief internet search for more information if needed. [TIP: Explore the help on the `summary()` function, in particular the `maxsum` argument...]

```
summary(Neonics$Species.Common.Name, maxsum = 7) #returned 7 since R created
```

```
##      Honey Bee      Parasitic Wasp Buff Tailed Bumblebee
##           667           285           183
## Carniolan Honey Bee      Bumble Bee      Italian Honeybee
##           152           140           113
##      (Other)
##           3083
```

```
#"Other" as a category
```

Answer: The 6 most commonly studied species are all bee species. Bees are pollinators. If they are harmed, our methods to produce food and other forms of vegetation are at risk.

8. Concentrations are always a numeric value. What is the class of `Conc.1..Author.` column in the dataset, and why is it not numeric? [Tip: Viewing the dataframe may be helpful...]

```
class(Neonics$Conc.1..Author.)
```

```
## [1] "factor"
```

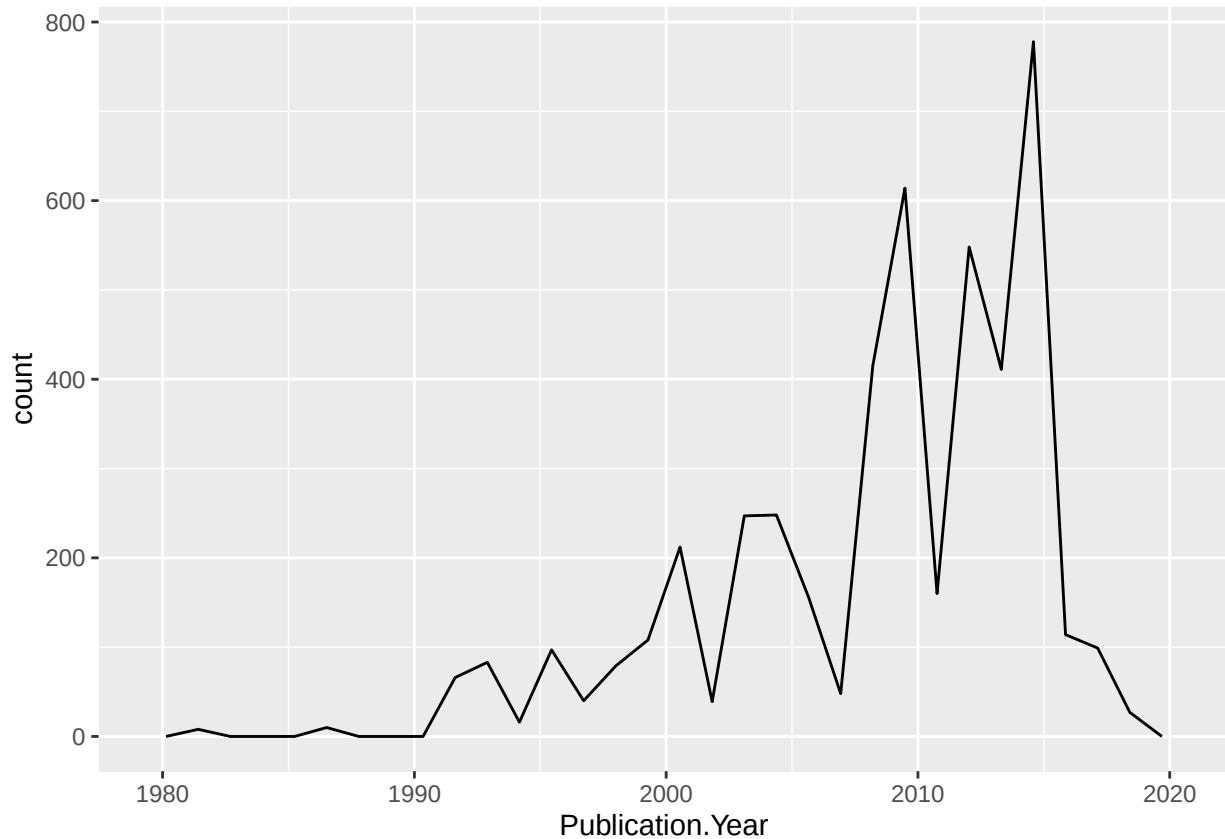
Answer: It is not numeric because the column includes inequalities (<,>,etc). Therefore, R can't read it as just one number

Explore your data graphically (Neonics)

9. Using `geom_freqpoly`, generate a plot of the number of studies conducted by publication year.

```
ggplot(Neonics) +  
  geom_freqpoly(aes(x = Publication.Year)) #publication year is the x axis and
```

```
## 'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.
```

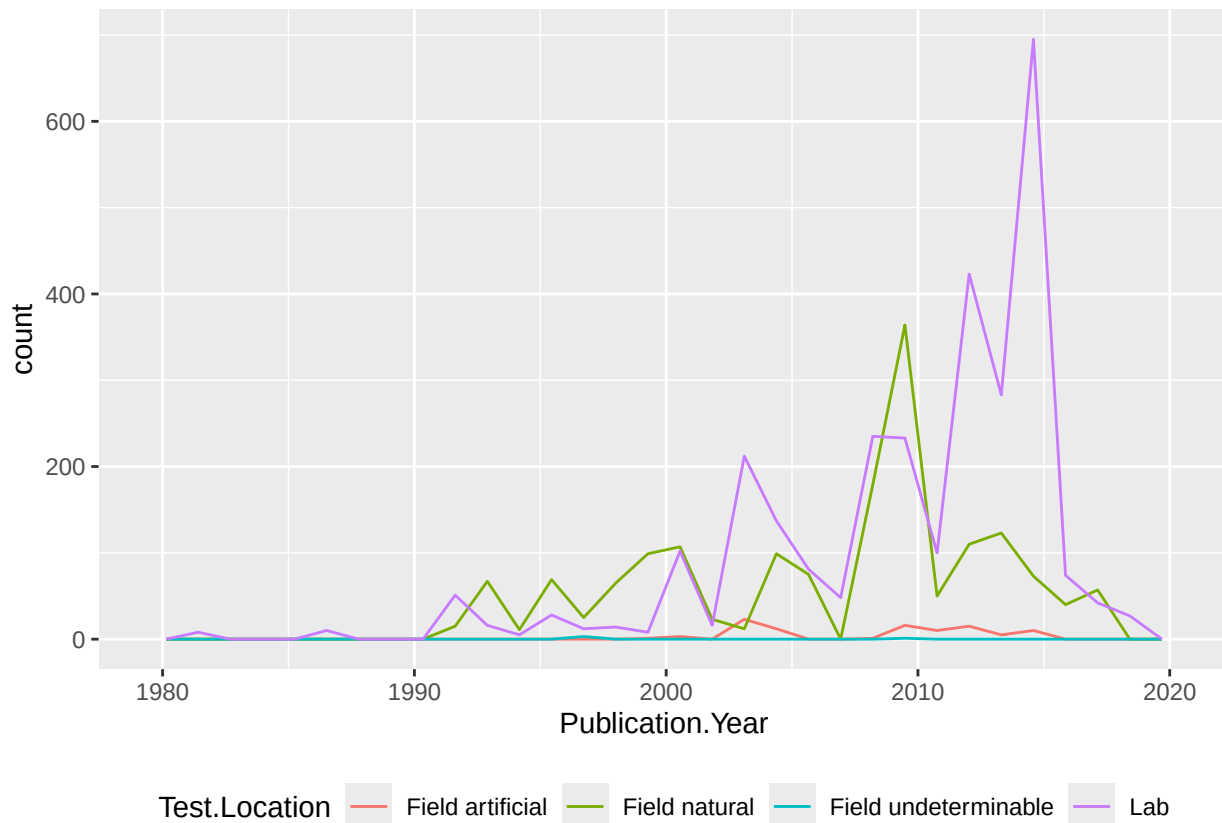


```
#number of publications is the y axis
```

10. Reproduce the same graph but now add a color aesthetic so that different Test.Location are displayed as different colors.

```
ggplot(Neonics) +
  geom_freqpoly(aes(x = Publication.Year, color = Test.Location)) +
  theme(legend.position = "bottom") #publication year is x axis, number of
```

'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.



#publications is y axis but now there are separate lines for each test location

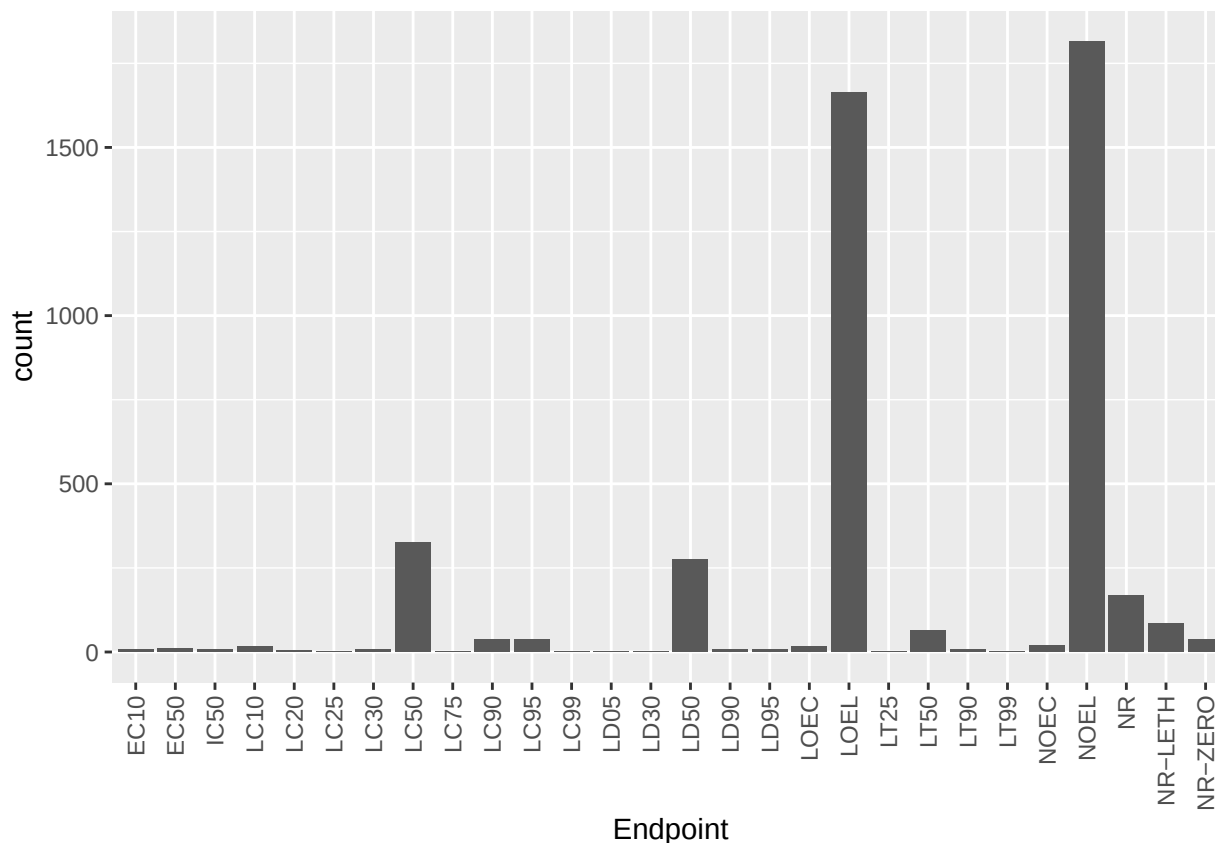
Interpret this graph. What are the most common test locations, and do they differ over time?

Answer: Field natural and field lab seem to go back and forth for most common test location between 1990 and 2010. Then, however, there's a clear drop off in the number of studies from "Field natural" test locations and a large spike in studies from "Lab" test locations.

11. Create a bar graph of Endpoint counts. What are the two most common end points, and how are they defined? Consult the ECOTOX_CodeAppendix for more information.

[TIP: Add `theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1))` to the end of your plot command to rotate and align the X-axis labels...]

```
ggplot(data = Neonic, aes(x = Endpoint)) +
  geom_bar() +
  theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1))
```



Answer: The most common endpoints are NOEL and LOEL. LOEL stands for Lowest-observable-effect-level. Basically is the lowest concentration (in this case of the insecticide) at which there is a significant effect. NOEL stands for No-observable-effect-level. It is the highest concentration (in this case of insecticide) at which there is no significant effect

Explore your data (Litter)

- Determine the class of collectDate. Is it a date? If not, change to a date and confirm the new class of the variable. Using the `unique` function, determine which dates litter was sampled in August 2018.

```
class(Litter$collectDate) #it is a factor, not a date
```

```
## [1] "factor"
```

```
Litter$collectDate = ymd(Litter$collectDate) #turns it to a date and tells R  
#that the factor is in the format year, month, day  
class(Litter$collectDate) #now it's class is a date
```

```
## [1] "Date"
```

```
unique(Litter$collectDate[format(Litter$collectDate, "%B %Y")=="August 2018"])
```

```
## [1] "2018-08-02" "2018-08-30"
```

```
#returns unique dates in August 2018
```

13. Using the `unique` function, list the different plotIDs sampled at Niwot Ridge. How is the information obtained from `unique` different from that obtained from `summary`?

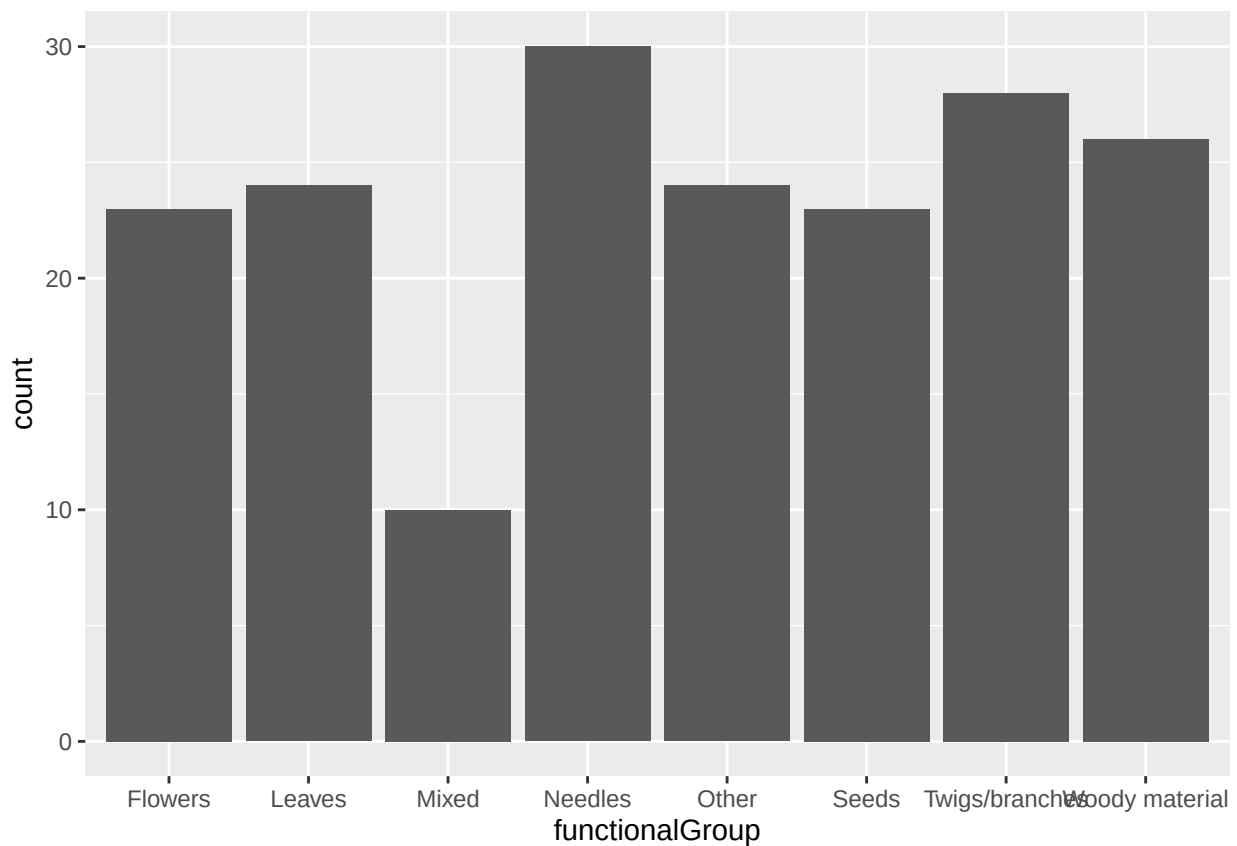
```
unique(Litter$plotID)
```

```
## [1] NIWO_061 NIWO_064 NIWO_067 NIWO_040 NIWO_041 NIWO_063 NIWO_047 NIWO_051  
## [9] NIWO_058 NIWO_046 NIWO_062 NIWO_057  
## 12 Levels: NIWO_040 NIWO_041 NIWO_046 NIWO_047 NIWO_051 NIWO_057 ... NIWO_067
```

Answer: `Summary()` tells us how many of each value are present while `unique()` just returns the unique values in a column

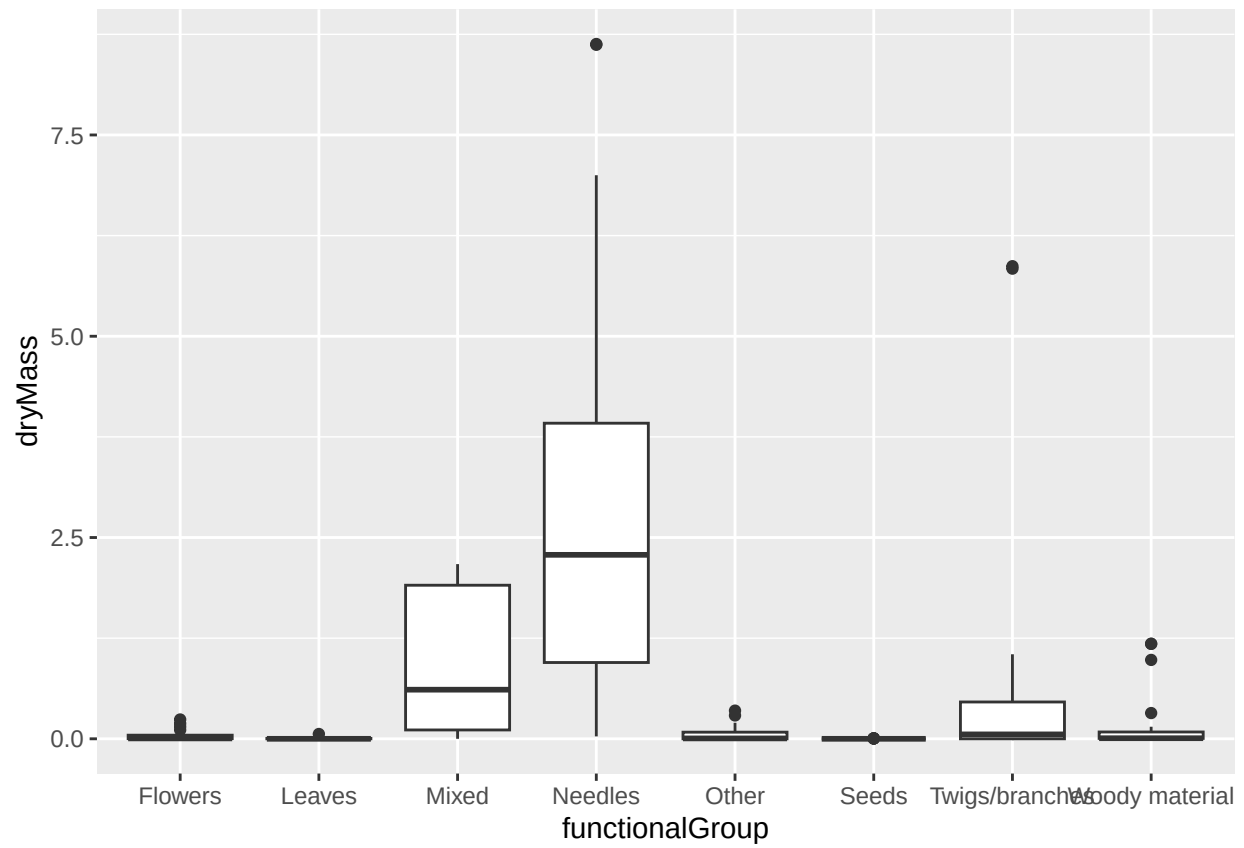
14. Create a bar graph of functionalGroup counts. This shows you what type of litter is collected at the Niwot Ridge sites. Notice that litter types are fairly equally distributed across the Niwot Ridge sites.

```
ggplot(data = Litter, aes(x = functionalGroup)) +  
  geom_bar()
```

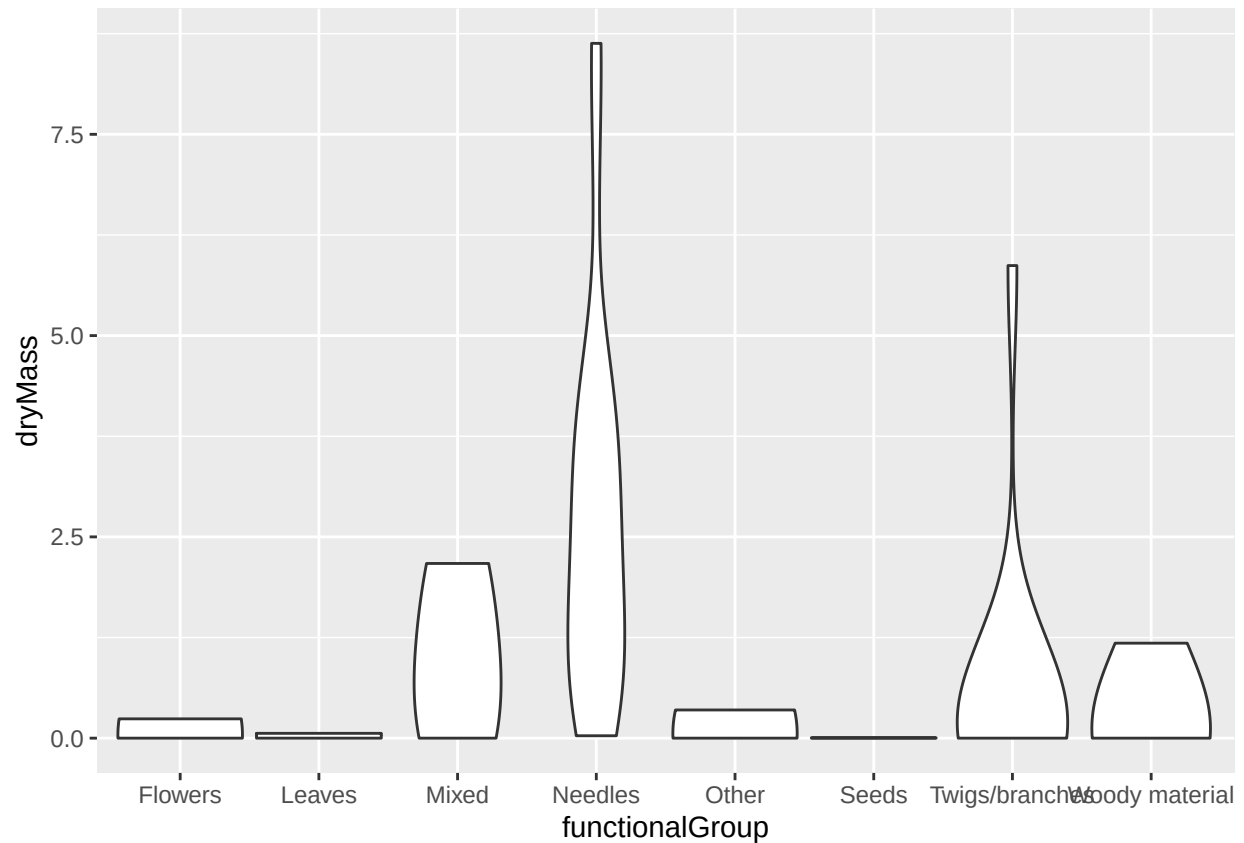


15. Using `geom_boxplot` and `geom_violin`, create a boxplot and a violin plot of `dryMass` by `functionalGroup`.

```
ggplot(Litter) +  
  geom_boxplot(aes(x = functionalGroup, y = dryMass))
```



```
ggplot(Litter) +  
  geom_violin(aes(x = functionalGroup, y = dryMass), bw = 1) #adjust the
```

#bandwidth of the lines so that we can actually see how the data is distributed

Why is the boxplot a more effective visualization option than the violin plot in this case?

Answer: There aren't many observations for each value within each group so without adjusting the bw, we just get lines. the boxplot shows us a better idea of the dry mass distribution within each functional group.

What type(s) of litter tend to have the highest biomass at these sites?

Answer: Needles seem to have the highest biomass at these sites.